

AI 3000: Reinforcement Learning

Tutorial 2 Solutions

16 August 2026

Note. The questions are reproduced in this solution sheet for convenience. Their notation (and some of the values) has been adjusted, where necessary, to agree with the notation used in the solutions.

1 One-Period MDP Problem

Consider the following one-Period MDP problem:

- States: $S = \{s_0, s_1, s_2\}$ with the state s_0 being the initial state, and s_1, s_2 being absorbing states.
- Actions: $A(s_0) = \{a_1, a_2, a_3, a_4, a_5\}$

and the one-step transition probabilities and rewards being defined as follows:

Action	$P(s_1 s_0, a)$	$P(s_2 s_0, a)$	$R(s_0, a, s_1)$	$R(s_0, a, s_2)$
a_1	0.6	0.4	+10	+0
a_2	0.5	0.5	+8	+0
a_3	0.9	0.1	+2	+0
a_4	0.2	0.8	+14	+4
a_5	0.3	0.7	+5	+2

1. Find the state-action values for all the available actions.
2. Find the set of all optimal deterministic policies for the setting given above. How many such policies exist?

Solution: One-Period MDP Problem

Because s_1 and s_2 are absorbing states, the process terminates after the transition from the initial state s_0 . Hence, for any action $a \in A(s_0)$, the state-action value is given by the expected immediate reward:

$$Q(s_0, a) = \sum_{s' \in \{s_1, s_2\}} P(s' | s_0, a) R(s_0, a, s').$$

1.1 State-Action Values

For action a_1 ,

$$Q(s_0, a_1) = 0.6(10) + 0.4(0) = 6.$$

For action a_2 ,

$$Q(s_0, a_2) = 0.5(8) + 0.5(0) = 4.$$

For action a_3 ,

$$Q(s_0, a_3) = 0.9(2) + 0.1(0) = 1.8.$$

For action a_4 ,

$$Q(s_0, a_4) = 0.2(14) + 0.8(4) = 6.$$

Finally, for action a_5 ,

$$Q(s_0, a_5) = 0.3(5) + 0.7(2) = 2.9.$$

Therefore, the state-action values are

a	a_1	a_2	a_3	a_4	a_5
$Q(s_0, a)$	6	4	1.8	6	2.9

In particular,

$$\max_{a \in A(s_0)} Q(s_0, a) = 6,$$

and the maximum is attained by both a_1 and a_4 .

1.2 Optimal Policies

Because s_0 is the only state at which an action is selected, a deterministic policy is completely characterized by its action at s_0 . A policy π is optimal if and only if

$$Q(s_0, \pi(s_0)) = \max_{a \in A(s_0)} Q(s_0, a) = 6.$$

From the state-action values computed above, this condition is satisfied precisely when

$$\pi(s_0) \in \{a_1, a_4\}.$$

Hence, the set of optimal deterministic policies is

$$\Pi^* = \{\pi^{(1)}, \pi^{(4)}\},$$

where

$$\pi^{(1)}(s_0) = a_1, \quad \pi^{(4)}(s_0) = a_4.$$

Thus, there are 2 optimal deterministic policies.

Remark. If randomized policies are also admitted, then there are infinitely many optimal policies. In fact, for any $\lambda \in [0, 1]$, consider the policy satisfying

$$\pi_\lambda(a_1 | s_0) = \lambda, \quad \pi_\lambda(a_4 | s_0) = 1 - \lambda,$$

and

$$\pi_\lambda(a_2 | s_0) = \pi_\lambda(a_3 | s_0) = \pi_\lambda(a_5 | s_0) = 0.$$

Its value is

$$\begin{aligned} v^{\pi_\lambda}(s_0) &= \lambda Q(s_0, a_1) + (1 - \lambda) Q(s_0, a_4) \\ &= \lambda(6) + (1 - \lambda)(6) \\ &= 6. \end{aligned}$$

Therefore, every such randomized policy is optimal. Under the usual interpretation of the question in terms of deterministic policies, however, there are exactly two optimal policies.

2 Finite Horizon MDPs

Consider a Finite-Horizon Markov Decision Process (MDP) defined by the following parameters:

- **State Space:** $S = \{s_1, s_2\}$
- **Action Spaces:** $A_{s_1} = \{a_{1,1}, a_{1,2}\}$ and $A_{s_2} = \{a_{2,1}, a_{2,2}\}$
- **Transition Probabilities:** For $n = 1, 2$:

$$\begin{aligned} p_n(s_1 | s_1, a_{1,1}) &= 0.8, & p_n(s_2 | s_1, a_{1,1}) &= 0.2 \\ p_n(s_1 | s_1, a_{1,2}) &= 0, & p_n(s_2 | s_1, a_{1,2}) &= 1.0 \\ p_n(s_1 | s_2, a_{2,1}) &= 0, & p_n(s_2 | s_2, a_{2,1}) &= 1.0 \\ p_n(s_1 | s_2, a_{2,2}) &= 0.4, & p_n(s_2 | s_2, a_{2,2}) &= 0.6 \end{aligned}$$

- **Rewards:** For $n = 1, 2$:

$$\begin{aligned} r_n(s_1, a_{1,1}, s_1) &= 5, & r_n(s_1, a_{1,1}, s_2) &= -5 \\ r_n(s_1, a_{1,2}, s_1) &= 0, & r_n(s_1, a_{1,2}, s_2) &= 5 \\ r_n(s_2, a_{2,1}, s_1) &= 0, & r_n(s_2, a_{2,1}, s_2) &= -5 \\ r_n(s_2, a_{2,2}, s_1) &= 20, & r_n(s_2, a_{2,2}, s_2) &= -10 \end{aligned}$$

Analyse the above MDP with horizon $N = 2$ and calculate $v^\pi(s_1) = \mathbb{E}^\pi[R_1 + R_2 | S_0 = s_1]$ for each of the following policies, where $S_0 = s_1$ is the initial state and R_n is the reward earned on the transition from S_{n-1} to S_n .

2.1 Markovian Deterministic Policy

Let $\pi_1 = (d_1, d_2)$ be a Markovian deterministic policy where the decision rules d_1 and d_2 are defined as follows:

$$d_1(s) = \begin{cases} a_{1,1}, & \text{if } s = s_1 \\ a_{2,1}, & \text{if } s = s_2 \end{cases} \quad \text{and} \quad d_2(s) = \begin{cases} a_{1,2}, & \text{if } s = s_1 \\ a_{2,1}, & \text{if } s = s_2 \end{cases}$$

2.2 Markovian Randomized Policy

Let $\pi_2 = (d_1, d_2)$ be a Markovian randomized policy where the decision rules are defined as follows:

$$w_{d_n}(a | s) = \begin{cases} q_1^n, & a = a_{1,1}, s = s_1 \\ 1 - q_1^n, & a = a_{1,2}, s = s_1 \\ q_2^n, & a = a_{2,1}, s = s_2 \\ 1 - q_2^n, & a = a_{2,2}, s = s_2 \end{cases}$$

where $q_1^1 = 0.1$, $q_2^1 = 0.5$, $q_1^2 = 0.5$, and $q_2^2 = 0.7$.

2.3 History-Dependent Deterministic Policy

Consider the policy $\pi_3 = (d_1, d_2)$, where the decision rules at each time-step are defined as follows:

$$d_1(h_1) = \begin{cases} a_{1,1}, & \text{if } h_1 = s_1 \\ a_{2,1}, & \text{if } h_1 = s_2 \end{cases} \quad \text{and} \quad d_2(h_2) = \begin{cases} a_{1,2}, & \text{if } h_2 = (s_1, a_{1,1}, s_1) \\ a_{2,1}, & \text{if } h_2 = (s_1, a_{1,1}, s_2) \\ a_{2,2}, & \text{if } h_2 = (s_2, a_{2,1}, s_2) \end{cases}$$

where h_i denotes the history available at decision step i .

2.4 History-Dependent Randomized Policy

The policy $\pi_4 = (d_1, d_2)$, where the decision rules at each time-step are defined as follows:

- At the first decision step, the randomized decision rule d_1 is given by:

$$w_{d_1}(a | h) = \begin{cases} q_1^1, & a = a_{1,1}, h = s_1 \\ 1 - q_1^1, & a = a_{1,2}, h = s_1 \\ q_2^1, & a = a_{2,1}, h = s_2 \\ 1 - q_2^1, & a = a_{2,2}, h = s_2 \end{cases}$$

where $q_1^1 = 0.1, q_2^1 = 0.5$

- For the second decision step, let the following be the possible histories:

$$\begin{aligned} h_1 &= (s_1, a_{1,1}, s_1), & h_4 &= (s_2, a_{2,1}, s_2) \\ h_2 &= (s_1, a_{1,1}, s_2), & h_5 &= (s_2, a_{2,2}, s_1) \\ h_3 &= (s_1, a_{1,2}, s_2), & h_6 &= (s_2, a_{2,2}, s_2) \end{aligned}$$

And the randomized decision rule d_2 depends on the history h_i as follows:

- For histories ending in state s_1 :

$$w_{d_2}(a | h_i) = \begin{cases} q_i^2, & a = a_{1,1} \\ 1 - q_i^2, & a = a_{1,2} \end{cases}$$

- For histories ending in state s_2 :

$$w_{d_2}(a | h_i) = \begin{cases} q_i^2, & a = a_{2,1} \\ 1 - q_i^2, & a = a_{2,2} \end{cases}$$

Here, $q_1^2 = 0.4, q_2^2 = 0.6, q_3^2 = 0.4, q_4^2 = 0.6, q_5^2 = 0.5$, and $q_6^2 = 0.5$.

Solution: Finite-Horizon MDPs

We consider the given finite-horizon MDP with horizon $N = 2$ and initial state $S_0 = s_1$. The total reward over the two stages is

$$R_1 + R_2,$$

where R_n denotes the reward obtained during the transition from S_{n-1} to S_n .

For a policy $\pi = (d_1, d_2)$, the expected total reward is therefore

$$v^\pi(s_1) = \mathbb{E}^\pi[R_1 + R_2 | S_0 = s_1].$$

We first evaluate the two Markovian policies analytically.

Evaluation of Markovian Policies

Let d_1 and d_2 denote the decision rules at the first and second stages, respectively. Because the policies are Markovian, the action chosen at each stage depends only on the current state.

The expected reward at the first stage is

$$\mathbb{E}^{d_1}[R_1 | S_0 = s_1] = \sum_{a^{(0)} \in A_{s_1}} w_{d_1}(a^{(0)} | s_1) \sum_{s^{(1)} \in S} p_1(s^{(1)} | s_1, a^{(0)}) r_1(s_1, a^{(0)}, s^{(1)}). \quad (1)$$

To evaluate the second-stage reward, we first determine the distribution of the intermediate state S_1 . For any $s^{(1)} \in S$,

$$\begin{aligned}\mathbb{P}(S_1 = s^{(1)} \mid S_0 = s_1) &= \sum_{a^{(0)} \in A_{s_1}} \mathbb{P}(A_0 = a^{(0)}, S_1 = s^{(1)} \mid S_0 = s_1) \\ &= \sum_{a^{(0)} \in A_{s_1}} w_{d_1}(a^{(0)} \mid s_1) p_1(s^{(1)} \mid s_1, a^{(0)}).\end{aligned}\quad (2)$$

Conditioned on the intermediate state $S_1 = s^{(1)}$, the expected reward at the second stage is

$$\begin{aligned}\mathbb{E}^{d_2}[R_2 \mid S_1 = s^{(1)}, S_0 = s_1] \\ = \sum_{a^{(1)} \in A_{s^{(1)}}} w_{d_2}(a^{(1)} \mid s^{(1)}) \sum_{s^{(2)} \in S} p_2(s^{(2)} \mid s^{(1)}, a^{(1)}) r_2(s^{(1)}, a^{(1)}, s^{(2)}).\end{aligned}\quad (3)$$

Consequently, by conditioning on the intermediate state,

$$\begin{aligned}\mathbb{E}^{d_2}[R_2 \mid S_0 = s_1] &= \mathbb{E}^{d_2}[\mathbb{E}^{d_2}[R_2 \mid S_1] \mid S_0 = s_1] \\ &= \sum_{s^{(1)} \in S} \mathbb{P}(S_1 = s^{(1)} \mid S_0 = s_1) \times \mathbb{E}^{d_2}[R_2 \mid S_1 = s^{(1)}, S_0 = s_1].\end{aligned}$$

Substituting from (2) and (3), we obtain

$$\begin{aligned}\mathbb{E}^{d_2}[R_2 \mid S_0 = s_1] &= \sum_{s^{(1)} \in S} \left[\sum_{a^{(0)} \in A_{s_1}} w_{d_1}(a^{(0)} \mid s_1) p_1(s^{(1)} \mid s_1, a^{(0)}) \right] \\ &\quad \times \left[\sum_{a^{(1)} \in A_{s^{(1)}}} w_{d_2}(a^{(1)} \mid s^{(1)}) \sum_{s^{(2)} \in S} p_2(s^{(2)} \mid s^{(1)}, a^{(1)}) r_2(s^{(1)}, a^{(1)}, s^{(2)}) \right].\end{aligned}\quad (4)$$

Hence, the expected total reward under a Markovian policy is

$$\boxed{v^\pi(s_1) = \mathbb{E}^{d_1}[R_1 \mid S_0 = s_1] + \sum_{s^{(1)} \in S} \mathbb{P}(S_1 = s^{(1)} \mid S_0 = s_1) \mathbb{E}^{d_2}[R_2 \mid S_1 = s^{(1)}, S_0 = s_1].} \quad (5)$$

This is the backward-induction decomposition of the two-stage value.

Transition Diagram

The possible transitions under the Markovian policies can be represented by the transition diagram 1.

We now evaluate the two Markovian policies separately.

Markovian Deterministic Policy

Consider the policy

$$\pi_1 = (d_1, d_2),$$

where

$$d_1(s) = \begin{cases} a_{1,1}, & s = s_1, \\ a_{2,1}, & s = s_2, \end{cases} \quad d_2(s) = \begin{cases} a_{1,2}, & s = s_1, \\ a_{2,1}, & s = s_2. \end{cases}$$

Because the initial state is $S_0 = s_1$, the first-stage action is deterministically $a_{1,1}$. Hence,

$$\begin{aligned}\mathbb{E}^{d_1}[R_1 \mid S_0 = s_1] &= \sum_{s^{(1)} \in S} p_1(s^{(1)} \mid s_1, a_{1,1}) r_1(s_1, a_{1,1}, s^{(1)}) \\ &= 0.8(5) + 0.2(-5) \\ &= 3.\end{aligned}$$

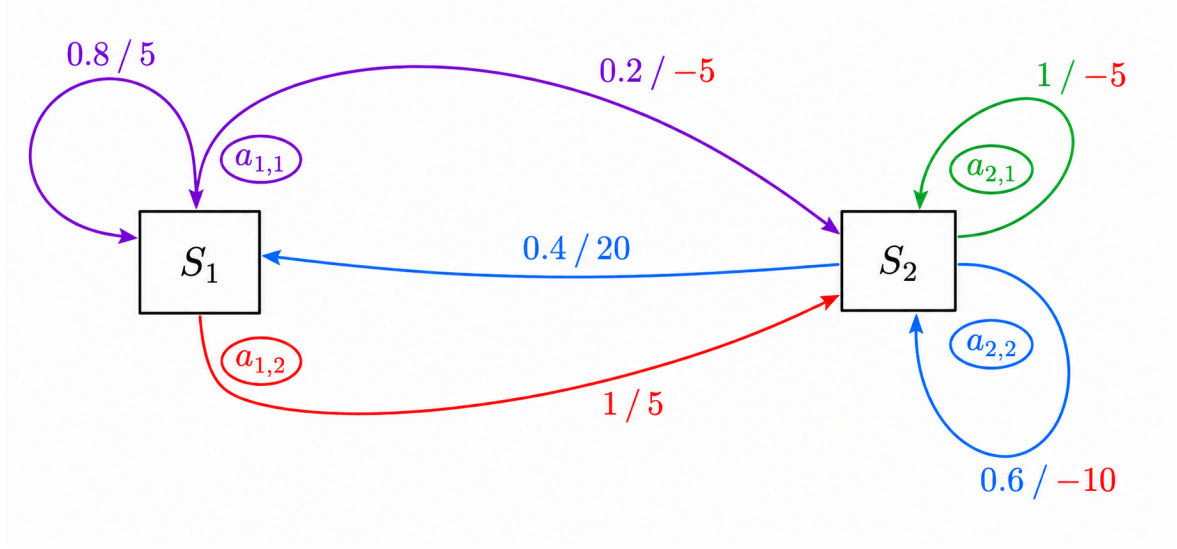


Figure 1: Transition diagram for the two-stage MDP.

The distribution of the intermediate state is

$$\mathbb{P}(S_1 = s_1 \mid S_0 = s_1) = 0.8, \quad \mathbb{P}(S_1 = s_2 \mid S_0 = s_1) = 0.2.$$

At the second stage, the action depends on the state S_1 .

If $S_1 = s_1$, the policy chooses $a_{1,2}$. Because

$$p_2(s_1 \mid s_1, a_{1,2}) = 0, \quad p_2(s_2 \mid s_1, a_{1,2}) = 1,$$

the expected second-stage reward is

$$\begin{aligned} \mathbb{E}^{d_2}[R_2 \mid S_1 = s_1] &= 0(0) + 1(5) \\ &= 5. \end{aligned}$$

If $S_1 = s_2$, the policy chooses $a_{2,1}$. Because

$$p_2(s_1 \mid s_2, a_{2,1}) = 0, \quad p_2(s_2 \mid s_2, a_{2,1}) = 1,$$

we obtain

$$\begin{aligned} \mathbb{E}^{d_2}[R_2 \mid S_1 = s_2] &= 0(0) + 1(-5) \\ &= -5. \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbb{E}^{d_2}[R_2 \mid S_0 = s_1] &= \mathbb{P}(S_1 = s_1 \mid S_0 = s_1)\mathbb{E}^{d_2}[R_2 \mid S_1 = s_1] \\ &\quad + \mathbb{P}(S_1 = s_2 \mid S_0 = s_1)\mathbb{E}^{d_2}[R_2 \mid S_1 = s_2] \\ &= 0.8(5) + 0.2(-5) \\ &= 3. \end{aligned}$$

Thus, the expected total reward under π_1 is

$$\begin{aligned} v^{\pi_1}(s_1) &= \mathbb{E}^{d_1}[R_1 \mid S_0 = s_1] + \mathbb{E}^{d_2}[R_2 \mid S_0 = s_1] \\ &= 3 + 3 \\ &= 6. \end{aligned}$$

2. Markovian Randomized Policy

Now consider the Markovian randomized policy

$$\pi_2 = (d_1, d_2),$$

with

$$q_1^1 = 0.1, \quad q_2^1 = 0.5, \quad q_1^2 = 0.5, \quad q_2^2 = 0.7.$$

At the first stage, because $S_0 = s_1$, the policy chooses $a_{1,1}$ with probability 0.1 and $a_{1,2}$ with probability 0.9.

Hence,

$$\begin{aligned} \mathbb{E}^{d_1}[R_1 \mid S_0 = s_1] &= 0.1(3) + 0.9(5) \\ &= 0.3 + 4.5 \\ &= 4.8. \end{aligned}$$

The distribution of S_1 is obtained by averaging the transition probabilities over the randomized first-stage action:

$$\begin{aligned} \mathbb{P}(S_1 = s_1 \mid S_0 = s_1) &= 0.1(0.8) + 0.9(0) \\ &= 0.08, \end{aligned}$$

and therefore

$$\begin{aligned} \mathbb{P}(S_1 = s_2 \mid S_0 = s_1) &= 0.1(0.2) + 0.9(1) \\ &= 0.92. \end{aligned}$$

At the second stage, if $S_1 = s_1$, the policy chooses $a_{1,1}$ with probability $q_1^2 = 0.5$ and $a_{1,2}$ with probability 0.5. Thus,

$$\begin{aligned} \mathbb{E}^{d_2}[R_2 \mid S_1 = s_1] &= 0.5[0.8(5) + 0.2(-5)] \\ &\quad + 0.5[0(0) + 1(5)] \\ &= 0.5(3) + 0.5(5) \\ &= 4. \end{aligned}$$

If $S_1 = s_2$, the policy chooses $a_{2,1}$ with probability $q_2^2 = 0.7$ and $a_{2,2}$ with probability 0.3.

For $a_{2,1}$, the expected reward is

$$0(0) + 1(-5) = -5.$$

For $a_{2,2}$, using

$$p_2(s_1 \mid s_2, a_{2,2}) = 0.4, \quad p_2(s_2 \mid s_2, a_{2,2}) = 0.6,$$

the expected reward is

$$0.4(20) + 0.6(-10) = 2.$$

Therefore,

$$\begin{aligned} \mathbb{E}^{d_2}[R_2 \mid S_1 = s_2] &= 0.7(-5) + 0.3(2) \\ &= -3.5 + 0.6 \\ &= -2.9. \end{aligned}$$

Consequently,

$$\begin{aligned} \mathbb{E}^{d_2}[R_2 \mid S_0 = s_1] &= 0.08(4) + 0.92(-2.9) \\ &= 0.32 - 2.668 \\ &= -2.348. \end{aligned}$$

Finally, the expected total reward under π_2 is

$$\begin{aligned} v^{\pi_2}(s_1) &= \mathbb{E}^{d_1}[R_1 \mid S_0 = s_1] + \mathbb{E}^{d_2}[R_2 \mid S_0 = s_1] \\ &= 4.8 - 2.348 \\ &= 2.452. \end{aligned}$$

Thus, the expected total rewards of the two Markovian policies are

$v^{\pi_1}(s_1) = 6, \quad v^{\pi_2}(s_1) = 2.452.$

Trajectory-Based Evaluation

The expected total reward can alternatively be computed by enumerating all feasible trajectories starting from $S_0 = s_1$. For each trajectory, its contribution to the expected reward is given by

$$\mathbb{P}(\text{trajectory}) \times (\text{total reward along the trajectory}).$$

1. Markovian Deterministic Policy

Consider

$$\pi_1 = (d_1, d_2),$$

where

$$d_1(s_1) = a_{1,1}, \quad d_1(s_2) = a_{2,1},$$

and

$$d_2(s_1) = a_{1,2}, \quad d_2(s_2) = a_{2,1}.$$

Because the policy is deterministic, the action probabilities are all equal to one along feasible trajectories.

The possible trajectories starting from $S_0 = s_1$ are shown below.

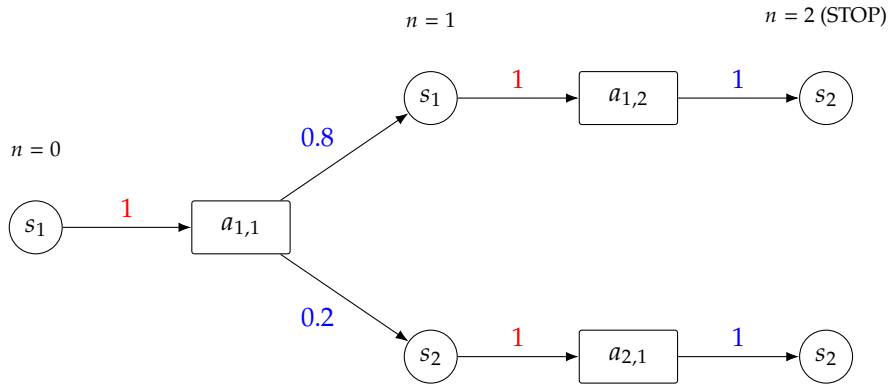


Figure 2: Feasible trajectories under the Markovian deterministic policy π_1 . Red labels denote policy probabilities and blue labels denote transition probabilities.

There are two feasible trajectories. Their probabilities and total rewards are

Trajectory	Probability	Total reward
$(s_1, a_{1,1}, s_1, a_{1,2}, s_2)$	$1(0.8)(1)(1) = 0.8$	$5 + 5 = 10$
$(s_1, a_{1,1}, s_2, a_{2,1}, s_2)$	$1(0.2)(1)(1) = 0.2$	$-5 - 5 = -10$

Hence,

$$\begin{aligned} v^{\pi_1}(s_1) &= 0.8(10) + 0.2(-10) \\ &= 8 - 2 \\ &= \boxed{6}. \end{aligned}$$

2. Markovian Randomized Policy

Now consider the Markovian randomized policy π_2 . At the first stage,

$$\pi_2(a_{1,1} | s_1) = 0.1, \quad \pi_2(a_{1,2} | s_1) = 0.9,$$

while at the second stage,

$$\begin{aligned} \pi_2(a_{1,1} | s_1) &= 0.5, & \pi_2(a_{1,2} | s_1) &= 0.5, \\ \pi_2(a_{2,1} | s_2) &= 0.7, & \pi_2(a_{2,2} | s_2) &= 0.3. \end{aligned}$$

The complete trajectory tree is shown below.

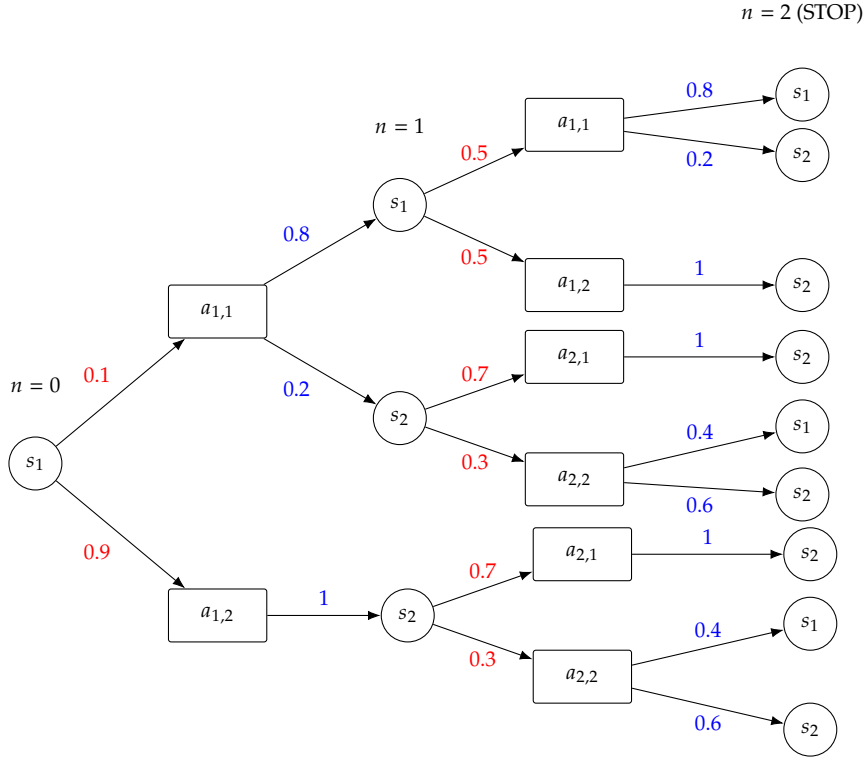


Figure 3: Feasible trajectories under the Markovian randomized policy π_2 . Red labels denote policy probabilities and blue labels denote transition probabilities.

For each trajectory, its probability is obtained by multiplying the policy probabilities and transition probabilities along the trajectory.

The resulting probabilities and rewards are

Trajectory	Probability	Total reward
$(s_1, a_{1,1}, s_1, a_{1,1}, s_1)$	$0.1(0.8)(0.5)(0.8) = 0.0320$	$5 + 5 = 10$
$(s_1, a_{1,1}, s_1, a_{1,1}, s_2)$	$0.1(0.8)(0.5)(0.2) = 0.0080$	$5 - 5 = 0$
$(s_1, a_{1,1}, s_1, a_{1,2}, s_2)$	$0.1(0.8)(0.5)(1) = 0.0400$	$5 + 5 = 10$
$(s_1, a_{1,1}, s_2, a_{2,1}, s_2)$	$0.1(0.2)(0.7)(1) = 0.0140$	$-5 - 5 = -10$
$(s_1, a_{1,1}, s_2, a_{2,2}, s_1)$	$0.1(0.2)(0.3)(0.4) = 0.0024$	$-5 + 20 = 15$
$(s_1, a_{1,1}, s_2, a_{2,2}, s_2)$	$0.1(0.2)(0.3)(0.6) = 0.0036$	$-5 - 10 = -15$
$(s_1, a_{1,2}, s_2, a_{2,1}, s_2)$	$0.9(1)(0.7)(1) = 0.6300$	$5 - 5 = 0$
$(s_1, a_{1,2}, s_2, a_{2,2}, s_1)$	$0.9(1)(0.3)(0.4) = 0.1080$	$5 + 20 = 25$
$(s_1, a_{1,2}, s_2, a_{2,2}, s_2)$	$0.9(1)(0.3)(0.6) = 0.1620$	$5 - 10 = -5$

Therefore,

$$\begin{aligned}
 v^{\pi_2}(s_1) &= 0.0320(10) + 0.0080(0) + 0.0400(10) + 0.0140(-10) \\
 &\quad + 0.0024(15) + 0.0036(-15) + 0.6300(0) \\
 &\quad + 0.1080(25) + 0.1620(-5) \\
 &= \boxed{2.452}.
 \end{aligned}$$

Thus, the trajectory-based calculation gives

$$v^{\pi_1}(s_1) = 6, \quad v^{\pi_2}(s_1) = 2.452,$$

which agrees with the analytical calculation.

Evaluation of History-Dependent Policies

For a history-dependent policy, the action selected at a given stage may depend on the entire history observed up to that stage. Consequently, the conditional distribution of the action at stage 2 cannot, in general, be determined solely from the intermediate state S_1 .

We therefore condition on the relevant histories when evaluating the second-stage reward.

3. History-Dependent Deterministic Policy

Consider the history-dependent deterministic policy

$$\pi_3 = (d_1, d_2),$$

where

$$d_1(h_1) = \begin{cases} a_{1,1}, & h_1 = s_1, \\ a_{2,1}, & h_1 = s_2, \end{cases}$$

and

$$d_2(h_2) = \begin{cases} a_{1,2}, & h_2 = (s_1, a_{1,1}, s_1), \\ a_{2,1}, & h_2 = (s_1, a_{1,1}, s_2), \\ a_{2,2}, & h_2 = (s_2, a_{2,1}, s_2). \end{cases}$$

Because the initial state is $S_0 = s_1$, the first-stage action is deterministically $a_{1,1}$. Thus,

$$\begin{aligned} \mathbb{E}^{d_1}[R_1 \mid S_0 = s_1] &= 0.8(5) + 0.2(-5) \\ &= 3. \end{aligned}$$

After the first transition, there are two possible histories:

$$h_1 = (s_1, a_{1,1}, s_1)$$

and

$$h_2 = (s_1, a_{1,1}, s_2).$$

Their probabilities are

$$\mathbb{P}(h_1 \mid S_0 = s_1) = 0.8, \quad \mathbb{P}(h_2 \mid S_0 = s_1) = 0.2.$$

If h_1 occurs, the second-stage action is $a_{1,2}$. Hence,

$$\begin{aligned} \mathbb{E}[R_2 \mid h_1] &= 0(0) + 1(5) \\ &= 5. \end{aligned}$$

If h_2 occurs, the second-stage action is $a_{2,1}$. Hence,

$$\begin{aligned} \mathbb{E}[R_2 \mid h_2] &= 0(0) + 1(-5) \\ &= -5. \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbb{E}^{d_2}[R_2 \mid S_0 = s_1] &= \mathbb{P}(h_1 \mid S_0 = s_1)\mathbb{E}[R_2 \mid h_1] \\ &\quad + \mathbb{P}(h_2 \mid S_0 = s_1)\mathbb{E}[R_2 \mid h_2] \\ &= 0.8(5) + 0.2(-5) \\ &= 3. \end{aligned}$$

Thus, the expected total reward is

$$\begin{aligned} v^{\pi_3}(s_1) &= \mathbb{E}^{d_1}[R_1 \mid S_0 = s_1] + \mathbb{E}^{d_2}[R_2 \mid S_0 = s_1] \\ &= 3 + 3 \\ &= \boxed{6}. \end{aligned}$$

Interestingly, for this particular policy, the history-dependent decision rule leads to the same expected value as the Markovian deterministic policy π_1 . The distinction between the two policies is nevertheless important: under π_3 , the second-stage action is specified as a function of the complete history rather than merely the current state.

4. History-Dependent Randomized Policy

Now consider the history-dependent randomized policy

$$\pi_4 = (d_1, d_2).$$

At the first stage, because $S_0 = s_1$,

$$\pi_4(a_{1,1} \mid s_1) = 0.1, \quad \pi_4(a_{1,2} \mid s_1) = 0.9.$$

Therefore, the expected first-stage reward is

$$\begin{aligned} \mathbb{E}^{d_1}[R_1 \mid S_0 = s_1] &= 0.1 [0.8(5) + 0.2(-5)] \\ &\quad + 0.9 [0(0) + 1(5)] \\ &= 0.1(3) + 0.9(5) \\ &= 4.8. \end{aligned}$$

At the second stage, only three of the specified histories can be reached from the initial state s_1 :

$$\begin{aligned} h_1 &= (s_1, a_{1,1}, s_1), \\ h_2 &= (s_1, a_{1,1}, s_2), \\ h_3 &= (s_1, a_{1,2}, s_2). \end{aligned}$$

The remaining histories

$$h_4 = (s_2, a_{2,1}, s_2), \quad h_5 = (s_2, a_{2,2}, s_1), \quad h_6 = (s_2, a_{2,2}, s_2)$$

cannot be reached from the specified initial state under the first-stage decision rule. Hence, their second-stage randomization probabilities do not affect $v^{\pi_4}(s_1)$.

The probabilities of the three reachable histories are

$$\begin{aligned} \mathbb{P}(h_1 \mid S_0 = s_1) &= 0.1(0.8) = 0.08, \\ \mathbb{P}(h_2 \mid S_0 = s_1) &= 0.1(0.2) = 0.02, \\ \mathbb{P}(h_3 \mid S_0 = s_1) &= 0.9(1) = 0.9. \end{aligned}$$

We now evaluate the expected second-stage reward conditional on each history.

For

$$h_1 = (s_1, a_{1,1}, s_1),$$

the policy selects $a_{1,1}$ with probability $q_1^2 = 0.4$ and $a_{1,2}$ with probability 0.6. Therefore,

$$\begin{aligned} \mathbb{E}[R_2 \mid h_1] &= 0.4 [0.8(5) + 0.2(-5)] \\ &\quad + 0.6 [0(0) + 1(5)] \\ &= 0.4(3) + 0.6(5) \\ &= 4.2. \end{aligned}$$

For

$$h_2 = (s_1, a_{1,1}, s_2),$$

the policy selects $a_{2,1}$ with probability $q_2^2 = 0.6$ and $a_{2,2}$ with probability 0.4. Thus,

$$\begin{aligned} \mathbb{E}[R_2 \mid h_2] &= 0.6 [0(0) + 1(-5)] \\ &\quad + 0.4 [0.4(20) + 0.6(-10)] \\ &= 0.6(-5) + 0.4(2) \\ &= -2.2. \end{aligned}$$

Finally, for

$$h_3 = (s_1, a_{1,2}, s_2),$$

the policy selects $a_{2,1}$ with probability $q_3^2 = 0.4$ and $a_{2,2}$ with probability 0.6. Hence,

$$\begin{aligned}\mathbb{E}[R_2 \mid h_3] &= 0.4[0(0) + 1(-5)] \\ &\quad + 0.6[0.4(20) + 0.6(-10)] \\ &= 0.4(-5) + 0.6(2) \\ &= -0.8.\end{aligned}$$

Combining the three possible histories gives

$$\begin{aligned}\mathbb{E}^{d_2}[R_2 \mid S_0 = s_1] &= 0.08(4.2) + 0.02(-2.2) + 0.9(-0.8) \\ &= 0.336 - 0.044 - 0.72 \\ &= -0.428.\end{aligned}$$

Therefore,

$$\begin{aligned}v^{\pi_4}(s_1) &= \mathbb{E}^{d_1}[R_1 \mid S_0 = s_1] + \mathbb{E}^{d_2}[R_2 \mid S_0 = s_1] \\ &= 4.8 - 0.428 \\ &= \boxed{4.372}.\end{aligned}$$

Thus, the analytical calculations for all four policies give

$$\boxed{v^{\pi_1}(s_1) = 6, \quad v^{\pi_2}(s_1) = 2.452, \quad v^{\pi_3}(s_1) = 6, \quad v^{\pi_4}(s_1) = 4.372.}$$

Trajectory-based calculation. We've left the path-by-path math for history-dependent policies (both the strict deterministic ones and the randomized) as an exercise! To tackle this, map out all the trajectories the system can take. For each path, figure out its overall probability and add up the total rewards. Finally, calculate the overall expected reward and make sure it perfectly matches the shortcut formulas we just covered above.

3 Policy-Value Function

Consider a simplified version of the Recycling Robot discussed in lecture 2, with the MDP defined as follows

- States: High(H) and Low(L)
- Actions: Search(s) and Recharge(r)

with the one-step transition probabilities and rewards being defined as follows:

- In State H :
 - Action Search: 80% chance to stay in H (Reward +10), 20% chance to drop to L (Reward +10).
 - Action Recharge: Waste of energy; stays in H (Reward 0).
- In State L :
 - Action Search: Risk of draining battery completely! 30% chance to stay in L (Reward +3), 70% chance battery dies (reset to H with penalty Reward -5).
 - Action Recharge: Deterministically moves back to H (Reward 0).

Now consider the following stationary policies:

1. Under π_1 , the robot always searches, i.e., $\pi_1(s \mid H) = \pi_1(s \mid L) = 1$.
2. Under π_2 , the robot is less risk-taking when in a low-charge state, i.e., $\pi_2(s \mid H) = 0.9$ and $\pi_2(s \mid L) = 0.4$.

Compare v^{π_1} and v^{π_2} under the expected discounted reward criterion with a two-step horizon for discount factors $\lambda = 0.1$ and $\lambda = 0.9$.

Solution: Policy-Value Function

For a stationary policy π and a two-step horizon, the value function under the discounted reward criterion is

$$v^\pi(s) = \mathbb{E}^\pi[R_1 \mid S_0 = s] + \lambda \mathbb{E}^\pi[R_2 \mid S_0 = s].$$

We fix λ and compare the two policies.

3.1 Always Search

Under π_1 , the expected one-step rewards are

$$r^{\pi_1}(H) = 10, \quad r^{\pi_1}(L) = 0.3(3) + 0.7(-5) = -2.6.$$

The corresponding transition matrix is

$$P^{\pi_1} = \begin{pmatrix} 0.8 & 0.2 \\ 0.7 & 0.3 \end{pmatrix}.$$

Thus,

$$\begin{aligned} v^{\pi_1}(H) &= 10 + \lambda [0.8(10) + 0.2(-2.6)] \\ &= \boxed{10 + 7.48\lambda}, \end{aligned}$$

and

$$\begin{aligned} v^{\pi_1}(L) &= -2.6 + \lambda [0.7(10) + 0.3(-2.6)] \\ &= \boxed{-2.6 + 6.22\lambda}. \end{aligned}$$

3.2 Less Risk-Taking in State L

Under π_2 ,

$$\pi_2(s \mid H) = 0.9, \quad \pi_2(s \mid L) = 0.4.$$

The expected one-step rewards are

$$r^{\pi_2}(H) = 0.9(10) = 9, \quad r^{\pi_2}(L) = 0.4(-2.6) = -1.04.$$

The policy-induced transition matrix is

$$P^{\pi_2} = \begin{pmatrix} 0.82 & 0.18 \\ 0.88 & 0.12 \end{pmatrix}.$$

Therefore,

$$\begin{aligned} v^{\pi_2}(H) &= 9 + \lambda [0.82(9) + 0.18(-1.04)] \\ &= \boxed{9 + 7.1928\lambda}, \end{aligned}$$

while

$$\begin{aligned} v^{\pi_2}(L) &= -1.04 + \lambda [0.88(9) + 0.12(-1.04)] \\ &= \boxed{-1.04 + 7.7952\lambda}. \end{aligned}$$

Hence, for a fixed λ ,

	H	L
π_1	$10 + 7.48\lambda$	$-2.6 + 6.22\lambda$
π_2	$9 + 7.1928\lambda$	$-1.04 + 7.7952\lambda$

The students can now substitute $\lambda = 0.1$ and $\lambda = 0.9$ and compare the two policies.

Exercise. How does changing λ affect the relative values of the two policies? In particular, why does a larger discount factor change the importance of the second-stage reward?

We will return to this example later and investigate this question computationally. In particular, we will use coding to study how the expected reward behaves for both finite- and infinite-horizon settings as the discount factor λ varies.